TASK 2

System Description

The implementation was developed and tested on a virtual network environment using **FreeBSD 12.0**. The topology consists of three Virtual Machines:

- **VM1 (Client)**: Acts as the source of traffic.
- **VM2 (Firewall Gateway)**: The central node where the custom kernel module is loaded to intercept packets via the `pfil(9)` framework.
- **VM3 (Web Server)**: The destination node hosting an Apache server at IP 192.168.2.20.

Commands to Execute and Test

To ensure a clean environment, all existing `pf` or `ipfw` rules were flushed before beginning.

1. Compilation and Loading (on VM2)

Execute the following to compile the module and load it into the kernel:

```
make
kldload ./simple_shield.ko
```

2. Monitoring Statistics (on VM2)

To view the real-time drop count and packet sizes, use the system message buffer:

```
dmesg | tail -f
```

3. Testing the Firewall (on VM1)

- **Normal Request**: `curl http://192.168.2.20`
  Expected: The server returns the webpage successfully.
- **Blocked Request**: `curl -H "Host:  blocked.com" http://192.168.2.20`
  Expected: The connection hangs or returns an empty reply, and VM2 logs a drop.

Assumptions

- **Protocol**: It is assumed the traffic is standard, unencrypted HTTP (Port 80) to allow the kernel to inspect the payload.
- **Deep Packet Inspection**: We assume the target domain string "blocked.com" appears within the first 1024 bytes of the TCP payload.
- **IP Forwarding**: It is assumed that `sysctl net.inet.ip.forwarding=1` is set on VM2 to allow transit traffic.